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Modes of Transmission, Personal Protective Equipment, and Isolation Precautions

Public Health and Safety EMS 313

Learning Objectives

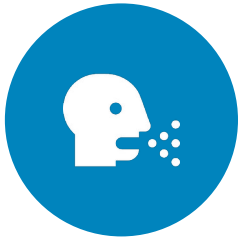
1. Describe the three routes of transmission: contact, droplet and airborne.
2. Explain how droplet and airborne spread differ, and why that changes the mask you wear.
3. State when PPE must be worn and removed, and choose gloves, gowns, masks and eye protection for a task.
4. Describe the sequence for putting on and taking off PPE.
5. Differentiate standard precautions from transmission-based precautions.
6. Describe contact, droplet and airborne precautions with examples of pathogens for each.
7. Differentiate isolation from quarantine.
8. Describe the precautions used for multidrug-resistant organisms.
9. Explain what administrators and education programmes must provide.
10. Apply the precautions to patients transported by ambulance.

Infections Spread by Three Routes



Contact

By touch, or through a surface



Droplet

Large drops that fall within two metres



Airborne

Tiny particles that stay in the air

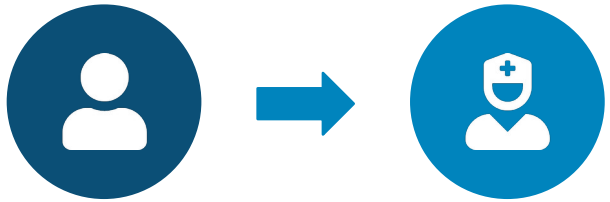


Our patient for the whole lecture: Ahmed, 34, from shared workers' accommodation. Cough for three weeks, fever, night sweats, weight loss. He coughs through a 25-minute transport.

Before you touch him: no mask, a surgical mask, or an N95?

Contact: By Touch, or Through Something in Between

Direct



Skin to skin. From the patient to you, and from you to the next patient.

Indirect

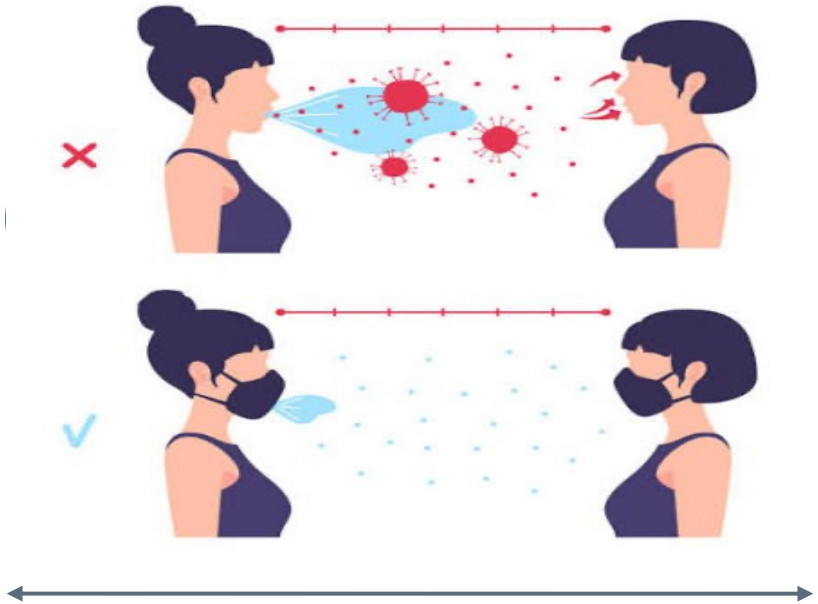


Through a stretcher rail, a cuff, a monitor cable, or a hand that touched them.

Indirect contact is the route that hand hygiene (Chapter 2) and equipment cleaning (Chapter 4) exist to break.

Ahmed: before he coughs at all, his hands, his blanket and the rail he holds are contact routes.

Droplets Are Large and Fall Within About Two Metres



about 2 metres, then they fall

Larger than 5 µm

Made by coughing, sneezing and talking. They travel a short distance and drop.

How far depends on the force of the cough, the density of the drops, temperature, humidity, air circulation, and how long the germ stays infectious.

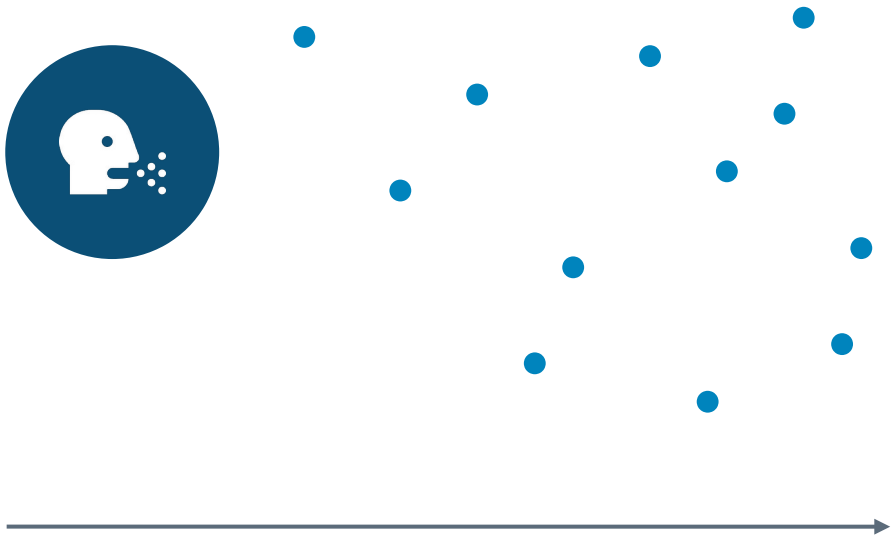
Example of infections

Pertussis (whooping cough), influenza, rhinovirus, SARS coronavirus, group A streptococcus, Neisseria meningitidis

Stand two metres away and droplets never reach you. In an ambulance you are never two metres away.

Ahmed: if this is influenza, his cough reaches everyone in the compartment. That alone needs a mask.

Airborne Particles Are Tiny, Stay in the Air, and Travel Far



across the room, along the corridor, for hours

5 µm or smaller: droplet nuclei

The dried residue of droplets. Light enough to float and move with the air.

Spread this way (Box 3.1)

Tuberculosis, measles (rubeola), chickenpox (varicella)

	Droplet	Airborne
Size	Larger than 5 µm	5 µm or smaller
Distance	About 2 metres, then falls	Far, stays suspended
Your mask	Surgical mask	N95 respirator, fit-tested

Ahmed: three weeks of cough, night sweats, weight loss. Think tuberculosis. He is airborne until proven otherwise, and a surgical mask is not enough.

PPE Is a Barrier Between the Germ and Your Skin, Airways and Clothes



Gloves



Gown



Mask



Eye protection

Put it on whenever blood or body fluids could reach you.

Take it off and throw it away before you leave the patient's room.

In the ambulance, the patient's room is the patient compartment. PPE comes off at the door, before the cab.

Putting On PPE: Gown, Mask, Eyes, Gloves



CDC sequence, from the source deck

1. **Gown**
   Cover arms and front, neck to mid-thigh. Tie at neck and waist.

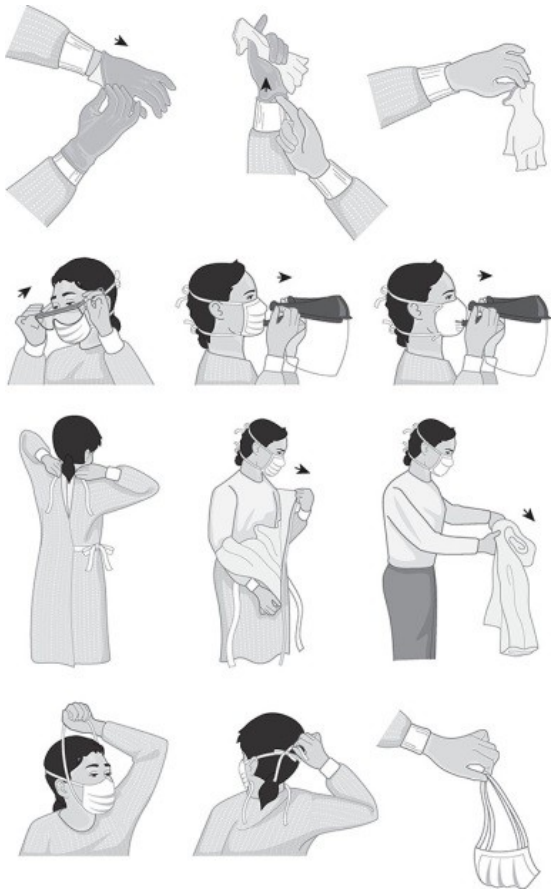
2. **Mask or respirator**
   Ties at the crown and the neck. Fit the nose piece. Seal-check a respirator.

3. **Goggles or face shield**
   Over the eyes, or the whole face.

4. **Gloves**
   Pull the cuffs over the gown sleeves.

Hand hygiene first, before the gown.

Taking Off PPE: The Front of Everything Is Contaminated



CDC sequence, from the source deck

1. **Gloves**
   Peel off without touching the outside.

2. **Goggles or face shield**
   By the headband or ear pieces.

3. **Gown**
   Untie, peel down from the shoulders, roll inside out.

4. **Mask or respirator**
   By the ties. Do not touch the front.

5. **Hand hygiene**
   Then, and only then, the cab and the report.

Clean parts

Ties, ear loops, headband: you may touch them with bare hands.

Contaminated parts

The front of the mask, the goggles, the shield, the gown, the gloves.

ACTIVITY · Spot the Mistake

After handover, Khalid pulls his mask down under his chin, removes his gloves, then his gown, and walks to the cab to write the report. Count the errors.

Gloves: Latex or Nitrile, Never Vinyl







Wear them whenever you could touch [IB/IC]

Blood or body fluids

Mucous membranes

Broken skin

Other infectious material

Gloves do not replace hand hygiene (Chapter 2). Hand hygiene before, hand hygiene after, a new pair for every patient.

Gowns: Full Coverage, for the Task, Off Before You Leave



An isolation gown, not a lab coat or jacket.

Covers the arms and the front, from the neck to mid-thigh. Always with gloves.

Wear it [IB/IC]

When the task could soil your clothes, and for direct contact with a patient whose secretions are not contained.

Two rules people miss

Do not gown just for entering a high-risk unit [IB]. Take the gown off and clean your hands before you leave the room [IB/IC].

Eyes, Nose and Mouth: Match the Protection to the Task



Choose a mask, goggles, a face shield, or a combination, for the task and the risk. [IB/IC]

Best eye protection: a face shield from the top of the head to the chin. It must be comfortable, adjustable, and not block your side vision (NIOSH).

Aerosol-generating procedures: full protection every time

Bronchoscopy

Suctioning the airway

Endotracheal intubation

If Ahmed needs suctioning or intubation on the way, the risk changes: respirator, full face shield, gown, gloves.

Three Kinds of Mask, and Only One Stops Airborne Germs



Surgical mask

Cleared by the FDA. Fluid-resistant.

Stops droplets and splashes.



Procedure or isolation mask

Not cleared by the FDA.

Droplet barrier, no fluid-resistance standard.



N95 respirator (the 'TB mask')

Filters tiny particles. Must be fit-tested.

The only one for airborne germs: TB, measles, chickenpox.

After use the front is contaminated. Remove it by the ties or ear loops, with bare hands, after the gloves are off.

Precautions

Two Tiers: Standard for Everyone, Extra for Some

Tier 1: Standard precautions

Every patient, every time.

Treat all blood, body fluids, secretions and excretions (except sweat), broken skin and mucous membranes as infectious

Tier 2: Transmission-based precautions

Contact, droplet or airborne.

Added when a patient has, or is suspected to have, a highly transmissible or important pathogen.

Tier 2 is added to Tier 1, never instead of it.

Ahmed: standard precautions from the first touch; airborne precautions added the moment the history is heard.

Contact Precautions: Gown and Gloves Before You Touch Anything



Single room if possible



Gown and gloves on entry, off before leaving



Equipment stays with the patient, or is cleaned between patients



Room cleaned at least daily

When: infection could spread by touching the patient or the surroundings

Draining wounds, faecal incontinence, or any discharge that could contaminate the environment. All measures are HICPAC/CDC Category IB.



Second patient: Umm Saad, 78, from a long-term care facility, with a draining leg wound. Her notes say MRSA. Gown and gloves before the stretcher, a cuff that stays with her, the ambulance cleaned before the next call, MRSA stated at handover.

Droplet Precautions: A Mask on You, a Mask on the Patient



Surgical mask on you, on entry [IB]



Surgical mask on the patient when moved, plus cough etiquette [IB]



Private room preferred



No special air handling; no respirator needed

For germs in droplets: pertussis, influenza, meningococcus, group A streptococcus

Spread needs close contact with respiratory secretions. The drops do not stay infectious over distance, so the air itself is not the problem.

The patient's mask matters as much as yours. It stops droplets at the source.

If Ahmed had influenza: surgical mask on the crew, surgical mask on Ahmed, and you would be done.

Airborne Precautions: A Respirator on You and a Room That Holds the Air



Fit-tested N95 or higher on you [IB]



Surgical mask on the patient until inside the room [IB/IC]



Negative-pressure isolation room (AIIR); air out through HEPA or to the outside



Non-immune staff stay away from measles, chickenpox, smallpox [IB]

For germs that stay in the air: tuberculosis, measles, chickenpox

No isolation room available: mask on the patient, and keep the room empty for about an hour afterwards [IB/IC].

The ambulance has no negative pressure. Mask on Ahmed, N95 on the crew, exhaust fan on, hospital pre-alerted, compartment aired after.

Which Precaution? Contact, Droplet or Airborne

1. A child with a barking cough and a whoop for two weeks
2. A man with fever, stiff neck and a purple rash
3. A care-home resident with faecal incontinence and a known VRE
4. A student with fever, cough and a rash that began on the face
5. An adult with fever, body aches and a dry cough in flu season
6. An adult with a month of cough and a friend recently treated for TB
7. A dialysis patient with a heavily draining catheter-site infection
8. A young adult with an itchy blistering rash all over the body

Answers in the speaker notes.

Isolation

Isolation Is for the Ill. Quarantine Is for the Exposed.



Isolation

The person is already ill or diagnosed. Usually in hospital, during treatment. Lasts as long as the patient is infectious.



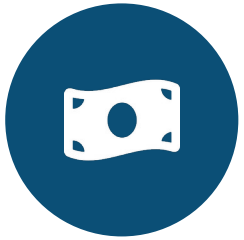
Quarantine

The person was exposed and may become ill. Used for a well-defined group and a dangerous, highly contagious disease. Contact with others is limited until it is clear they did not catch it.

After the call: Ahmed's tuberculosis is confirmed. He is isolated in the hospital's negative-pressure room until he is no longer infectious. The crew were exposed; they are not ill, so they are not isolated. They are screened and followed up.

System

None of This Works Unless the Service Pays for It and Teaches It



Administrators [IB/IC]

Put prevention of transmission into the safety objectives. Make it a priority, with the money and the people to run the programme.



Education on PPE and precautions

Task-specific. At orientation, then updated. For everyone. Built on adult learning. In a language and reading level staff understand.

Fit-testing, respirators in every vehicle, face shields at the stretcher, gowns that fit, someone whose job it is to check. These are administrative decisions.

The Whole Chapter in Five Questions

1	**How does it spread?**	Contact, droplet or airborne. The route decides everything else.
2	**What barrier do I need?**	Gloves, gown, mask or respirator, eye protection. Chosen for the task, removed in order.
3	**Which precaution?**	Standard for everyone. Contact, droplet or airborne added on suspicion.
4	**Who is separated from whom?**	Isolation for the ill, quarantine for the exposed, contact precautions for resistant organisms.
5	**What must the service provide?**	Objectives, budget, equipment, fit-testing and training for everyone.

“The precaution starts on suspicion, not on diagnosis.”

References

McGuire-Wolfe C. Foundations of Infection Control and Prevention. Burlington, MA: Jones & Bartlett Learning; 2018. Chapter 3, Modes of Transmission, Personal Protective Equipment, and Isolation Precautions.

Siegel JD, Rhinehart E, Jackson M, Chiarello L, Healthcare Infection Control Practices Advisory Committee. 2007 Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings. Atlanta, GA: CDC; 2007.

Source lecture: Chapter 3, Modes of Transmission, Personal Protection Equipment, and Isolation Precautions. Jones & Bartlett Learning, an Ascend Learning Company; 2018 (course slide deck, EMS 313). The PPE sequence figures on slides 8 and 9 are the CDC illustrations reproduced in that deck.